HANDS-ON ACTIVITY FOCUS GUIDE

Be the Control Loop

Check, read, command, report, and repeat

22 MIN ACTIVITY	8 STUDENT ROLES	UNPLUGGED FIRST FORMAT	1 SAFETY LEAD

ACTIVITY OUTCOME Students trace complete while-loop iterations, use fresh inputs on each cycle, and exit immediately when the active condition becomes false.

Materials

- Printed BasicTwoMotorTeleOp.java with space to mark braces.
- Role cards: CHECK, READ, COMMAND, REPORT, DRIVER, CODE TRACER, and STOP.
- Program token; TRUE/FALSE cards; three left/right joystick-value cards.
- Optional secured robot and telemetry display for the final extension.
- POWERING ON safety poster if the live extension will be used.

Before students enter

- Mark the while-loop opening and closing braces on your teacher copy.
- Prepare input pairs such as +0.4/+0.4, +0.6/+0.2, and 0.0/0.0.
- Arrange CHECK -> READ -> COMMAND -> REPORT in a loop with STOP outside it.
- Keep the live robot disabled; the core activity requires no motor power.

SAFETY BOUNDARY Students never cross into the robot test area while the OpMode is enabled. STOP and visually confirm DISABLED before anyone approaches.

Student setup and roles

Assign one primary role to each student. Rotate roles on a second trial if time permits. The safety lead may stop the activity at any time.

STUDENT	ROLE	RESPONSIBILITY
1	CHECK	Reads opModeIsActive() and chooses repeat or exit.
2	READ	Accepts the newest joystick-value card.
3	COMMAND	Sends the current values to the motors.
4	REPORT	Announces the telemetry values.
5	Driver	Changes the input card between cycles.
6	Code tracer	Points to each represented code line.
7	STOP	Receives the token when the condition is false.
8	Recorder	Documents each completed iteration.

Ground rules

- Predict before testing; record evidence before proposing a correction.
- Only the Driver Station operator touches INIT, PLAY, or STOP.
- Any student may call STOP for unexpected motion, instability, heat, noise, or an unclear boundary.
- Change only one variable between tests so the evidence remains interpretable.

POWERING ON gate

SAY AND DO ALL SIX STEPS 1. Secure the robot. 2. Confirm DISABLED. 3. Look around. 4. Make sure everyone is clear. 5. Call POWERING ON! and wait. 6. Visually confirm clear, then enable/PLAY.

After every test: press STOP, visually confirm DISABLED, and only then permit anyone to approach the robot.

22-minute teacher walkthrough

1. Find the loop boundary | 4 minutes

TEACHER	*"Ask students to box the condition and connect the opening and closing braces of the while statement."*

STUDENT	*"The code tracer outlines exactly which statements belong to the loop body."*

ASK	*"Do the braces repeat?"*

LISTEN FOR The statements inside the braces repeat; the condition is checked before each attempted iteration.

TEACHING NOTE Students often include the closing code or miss telemetry. Make them trace brace to matching brace.

2. Compare if and while | 4 minutes

TEACHER	*"Point to the outer if and inner while. Ask how many times each condition is evaluated when execution reaches it."*

STUDENT	*"CHECK demonstrates: if chooses once; while checks again before every cycle."*

ASK	*"Why is the outer if redundant here?"*

LISTEN FOR The while checks the same active condition before entering and before every repetition.

TEACHING NOTE Keep the outer if in this source walkthrough; refactoring can come after students understand the behavior.

3. Trace three fresh cycles | 8 minutes

TEACHER	*"Give the driver a different input pair before each cycle. Pass the token CHECK -> READ -> COMMAND -> REPORT -> CHECK."*

STUDENT	*"Students announce condition, input, motor command, and telemetry. The recorder completes one row per cycle."*

ASK	*"Why must READ happen again?"*

LISTEN FOR The driver's hands may have moved; responsive TeleOp uses the newest gamepad values.

TEACHING NOTE Do not let the token return to IDENTIFY, CONNECT, or WAIT. Those startup actions do not repeat.

4. Exit on FALSE | 3 minutes

TEACHER	*"Replace TRUE with FALSE when the token returns to CHECK. Ask what happens before another READ."*

STUDENT	*"CHECK passes the token directly to STOP. READ must not accept another input card."*

ASK	*"Does the loop finish one more cycle after FALSE?"*

LISTEN FOR No. The condition is checked first, so the body does not execute again.

TEACHING NOTE The immediate stillness after STOP makes the exit condition concrete.

5. Optional telemetry transfer | 3 minutes

TEACHER	*"If using the robot, secure it and run the complete POWERING ON sequence before PLAY. Keep sticks centered and observe telemetry changing with a gentle input."*

STUDENT	*"Students name the human role represented by each changing value, then the operator presses STOP and confirms DISABLED."*

ASK	*"Which role produces the displayed values?"*

LISTEN FOR REPORT/telemetry displays the values computed and commanded during that cycle.

TEACHING NOTE Skip live power if time, staffing, or robot readiness is not adequate.

Control-loop cycle record

Student/team: ________________________________ Date: ______________ Robot: __________________

CYCLE	ACTIVE?	NEW INPUT L / R	COMMAND / REPORT / NEXT
1	TRUE	+0.4 / +0.4	Motors: ____ Telemetry: ____ Next: ____
2	TRUE	+0.6 / +0.2	Motors: ____ Telemetry: ____ Next: ____
3	TRUE	0.0 / 0.0	Motors: ____ Telemetry: ____ Next: ____
4	FALSE	New card withheld	Does loop body run? ____ Next: ____
Student-created	____	____ / ____	Prediction: _________________________

STUDENT EVIDENCE RULE Write what was commanded and what was observed. Do not write a correction until STOP has been pressed and DISABLED has been confirmed.

Safety sign-off

Before live testing, the safety lead confirms: robot secure [ ] DISABLED [ ] boundary clear [ ] POWERING ON called and wait completed [ ]

After testing: STOP pressed [ ] DISABLED visually confirmed [ ] Safety lead initials: __________

COMPLETION CRITERIA The student uses lesson vocabulary, predicts before testing, records evidence, explains one result from the code or math, and follows STOP/DISABLED rules before approaching the robot.

Debrief

ASK	LISTEN FOR
Which jobs repeat?	CHECK, READ, COMMAND, and REPORT repeat while the condition remains true.
Why is an active check placed before the body?	It prevents another body execution after STOP or another stop request.
What would happen if inputs were read only once?	The robot would keep using stale values and feel unresponsive to later stick movement.
Does motor power 0.0 end the loop?	No. It requests stopped motors while the OpMode can remain active.

Troubleshooting without guessing

OBSERVATION	NEXT CHECK AFTER STOP/DISABLED
Token returns to startup	Redirect REPORT to CHECK, then TRUE sends it to READ.
Students skip telemetry	Require REPORT before the token can return to CHECK.
FALSE still performs READ	Place FALSE physically between CHECK and STOP; body actions are bypassed.
Live values appear unchanged	Confirm the gamepad is connected and that telemetry.update() occurs inside the loop.

Technical notes for the teacher

- opModeIsActive() becomes true after START and remains true until a stop is requested.
- A while condition is evaluated before each attempted iteration.
- The supplied outer if is redundant because the while checks the same condition.
- The loop has no explicit delay and runs as quickly as its operations and runtime permit.
- Fresh gamepad reads on every iteration are what make driver control responsive.

EXIT TICKET Describe one full loop cycle and explain exactly what prevents the next cycle after STOP.